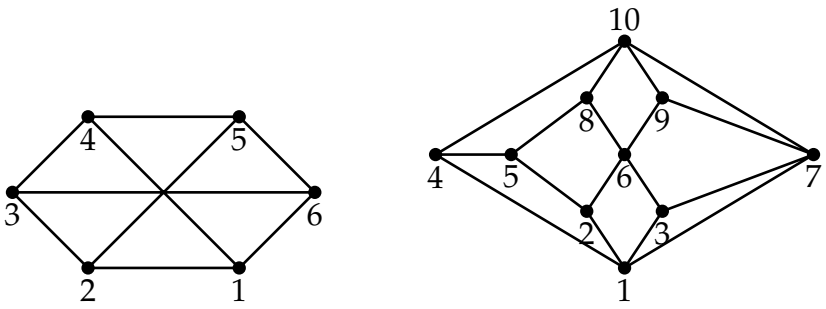
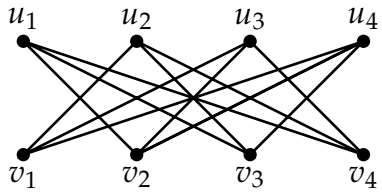
Assignment 2

Due: Monday 02 November, 2026.

1. For any graph G, show the following inequalities:
   (a) $\beta_0(G) + \beta_0(G^c) \geq \alpha_0(G) + \alpha_0(G^c)$.
   (b) $\beta_0(G) \geq \omega(G)$,
   where $\omega(G) = \max\{|V(H)| : H \text{ is a complete subgraph of } G\}$ is the clique number of G.
   (c) $\beta_0(G) \geq \delta(G)$.
2. Let M, N be disjoint matchings of a graph G with $|M| > |N|$. Show that there are disjoint matchings M' and N' of G such that $|M'| = |M| - 1$ and $|N'| = |N| + 1$ and $M' \cup N' = M \cup N$.
3. Illustrate the greedy algorithm to get a maximal independent set for the simple graph whose vertices are a, b, c, d, e, f and whose edges are (a, b), (b, c), (c, d), (d, e), (e, f), (f, a), (b, f), (c, e).
4. Apply greedy algorithm to get a maximal matching and a vertex cover of the graphs shown below.



5. Order the vertices of the graph shown below, in such a way that the greedy algorithm uses 4 colors.



6. Let G be the graph 3-cube Q_3. Order the vertices of G in 3 different ways, say S_1, S_2 and S_3 so that the greedy algorithm
   (a) uses 2 colors, if S_1 is the input,
   (b) uses 3 colors, if S_2 is the input, and
   (c) uses 4 colors, if S_3 is the input.

7. Find the edge-chromatic number of the following graphs.

   (a) $T_k + C_{2k+1}$, where T_k is a tree on k vertices.

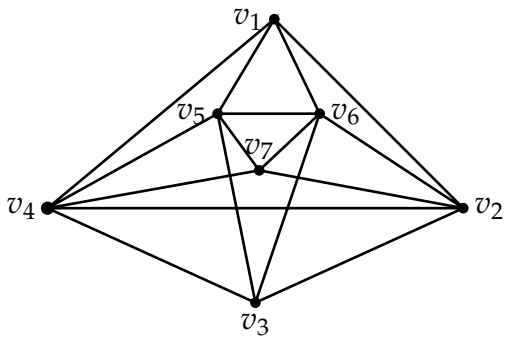
   (b) $C_5 + C_5$.

   (c) $C_{2s} + C_{2t}$.

8. In a school there are seven vacancies, one each in the departments of Chemistry, English, French, Geography, History, Mathematics and Physics. There are seven applicants $A_1, \ldots, A_7$ for the vacancies. The applicants and their specializations are listed in the table below. Determine the maximum number of suitably qualified teachers the school can employ. Justify your claim.

Teachers	Specialization
A_1	Mathematics, Physics
A_2	Chemistry, English, Mathematics
A_3	Chemistry, French, History, Physics
A_4	English, French, History, Physics
A_5	Chemistry, Mathematics
A_6	Mathematics, Physics
A_7	English, Geography, History.

9. Consider a graph G with n vertices and m edges, where n is odd. If $2m > (n-1)\Delta(G)$, then prove that $\chi_1(G) - 1 = \Delta(G)$. Hence, compute $\chi_1(G)$ for the graph G in the Figure shown below. Also label the edges of G with $\chi_1(G)$ colors.



10. (a) Show that there exists no 3-regular simple planar graph on 10 vertices whose girth is 5.

    (b) Draw a plane 3-regular simple graph on 10 vertices whose girth is 4.

11. Let G be a simple plane connected 3-regular graph with every face having degree 5 or 6. If p is the number of pentagonal faces and h is the number of hexagonal faces in G, show that $p = 12$ and $n(G) = 20 + 2h$.

12. Prove that the complement of simple planar graph with at least 11 vertices is nonplanar. Construct a self-complementary simple planar graph with 8 vertices.